Advanced Programming Practice · 21CSC203P

Unit 3 · Session 9 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 3 : Session â€“ 9 : SLO â€“ 1 | Describe MVC Architecture in GUI Development

I. Fill in the Blanks

1. View â€” the View is responsible for presenting the user interface.
2. Controller â€” it listens to user actions and modifies the Model.
3. Controller â€” an ActionListener attached to a button plays the Controller role in Swing MVC.
4. reusability â€ MVC promotes code reusability (separation of concerns) and testability.

II. True or False

1. FALSE â€ the Controller updates the Model; the View is refreshed from the Model, not updated directly by the Controller bypassing it.
2. TRUE â€ MVC separates data (Model), presentation (View) and input handling (Controller), which is the separation of concerns.

III. Multiple Choice

1. C â€ the Model contains and manages the application data and business logic.
2. C â€ the MVC control flow is View â†' Controller â†' Model â†' View.

IV. Programming

1. Identify the output: A â€ each button click increments the counter in the Model and the label is updated with the new value.
2. Identify the missing part: C â€ view.updateDisplay(model.getData()); after the Model changes, the View must be refreshed from the Model to complete the MVC cycle.